

Course Name: Switchgear and Protection Sessional
Course Code: EEE 4204

Line to Line Fault Detection with Temperature Protection



Presented To

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INTRODUCTION:

A line-to-line (L-L) fault is a type of short-circuit fault in a power system where two phases (conductors) come into direct contact with each other.

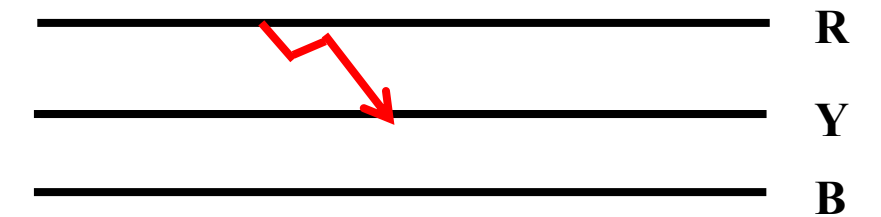
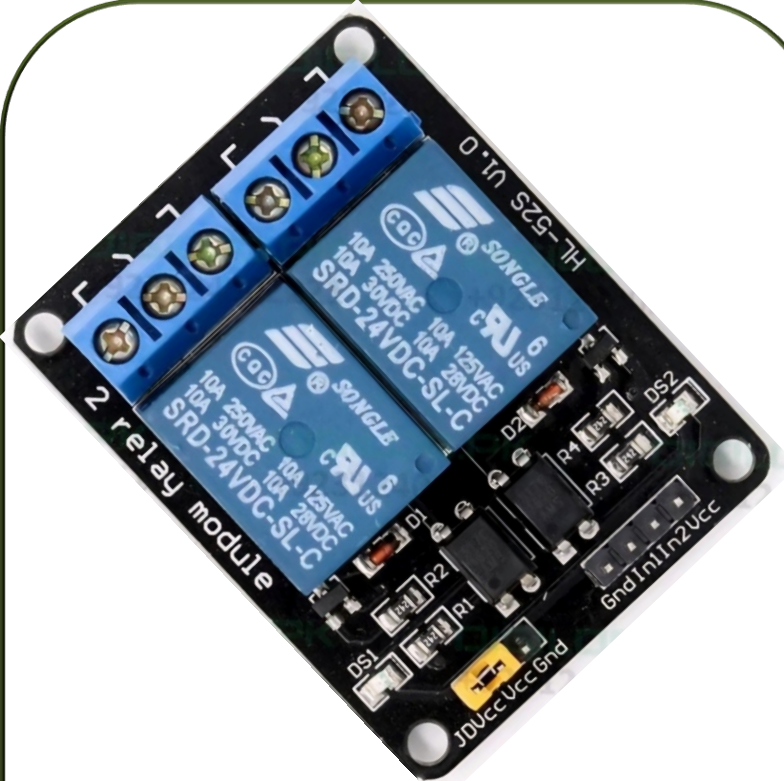


Fig-1: Line to Line Fault

Line-to-line faults can occur due to:

- Insulation failure between conductors.
- Mechanical damage to transmission/distribution lines (storms, falling trees, strong winds).
- Overvoltage (lightning strike or switching surge).
- Birds, animals, or foreign objects bridging two phases.
- Ageing of equipment (weakening insulation, loose connections).

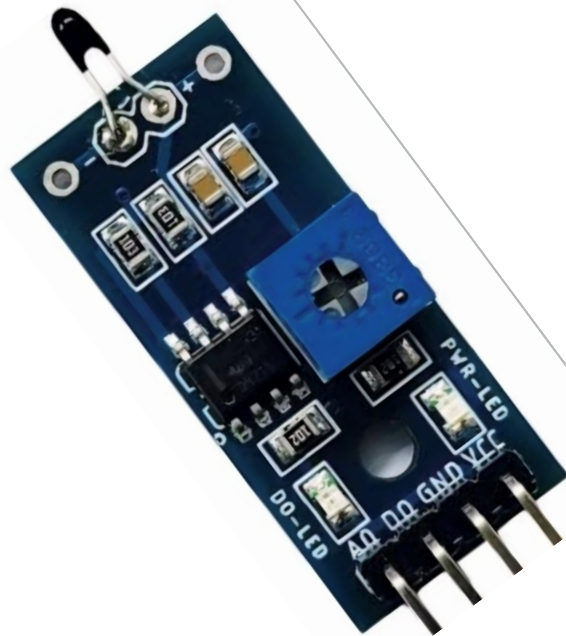
REQUIRED MATERIALS:



2 Channel 5V Relay Board Module



AC to DC Converter



NTC Thermistor Temperature Sensor Module



Bulb

- ☐ Connecting wire.
- ☐ Holder
- ☐ Plug

CONNECTION DIAGRAM:

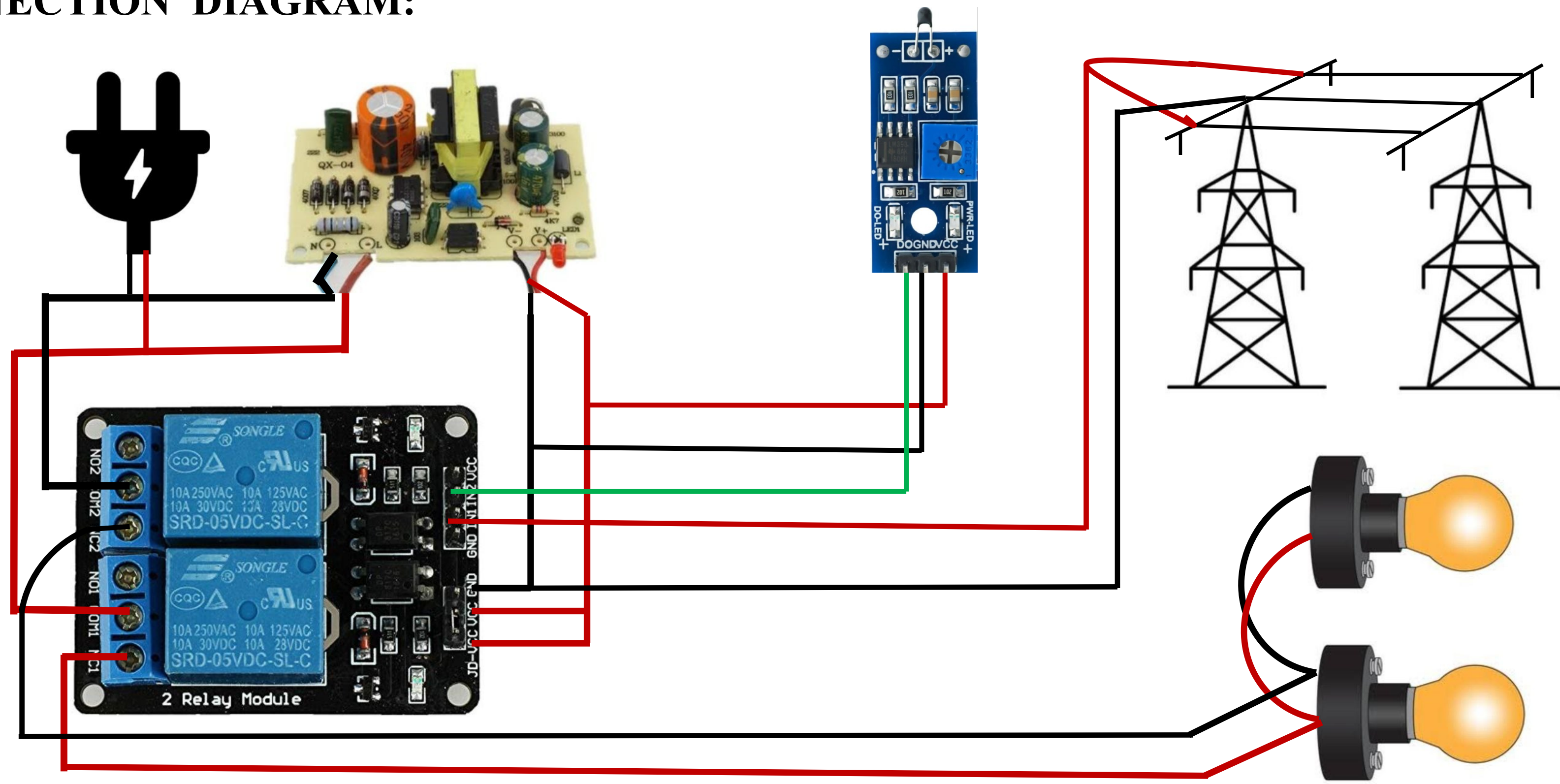
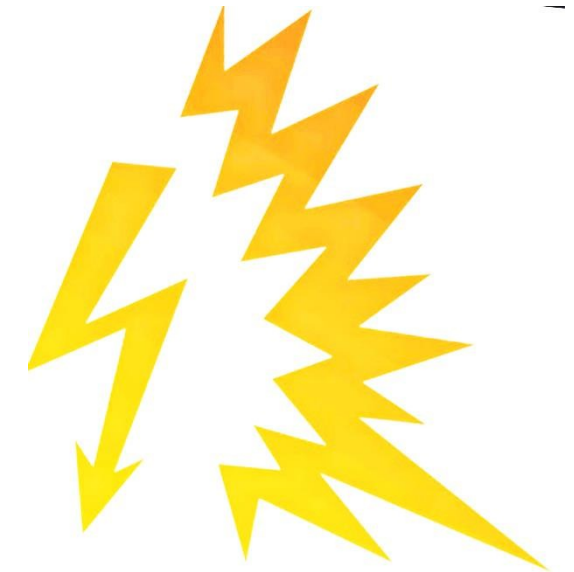


Fig-2: Line to Line Fault Detection with Temperature Protection.

PURPOSE:

The purpose of L–L fault detection is to:

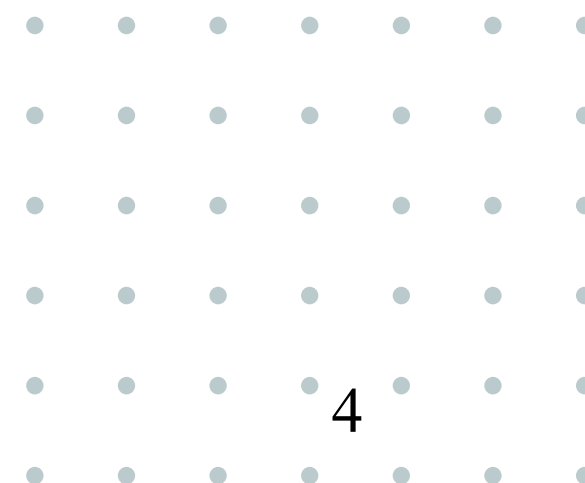
- Protect the system from damage caused by high fault current.
- Isolate the faulty section quickly to keep the rest of the system running.
- Prevent fire hazards and equipment failure.
- Maintain stability of the power system.



APPLICATIONS:

Fault detection systems are applied in:

- Transmission lines – to avoid large-scale blackouts.
- Distribution systems – to protect feeders and consumers.
- Transformers & Generators – to save windings from overheating/damage.
- Switchgear & Substations – to trip breakers and clear faults fast.
- Industrial plants – to protect motors and sensitive machines.



THANK YOU

